

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

Code of Conduct:

Iharish kumar .M/192565053 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

harish kumar .M

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Answer:

Cost — correct. Star needs a hub/switch plus a dedicated cable run to every node, which adds hardware and cabling cost. Bus just needs one shared backbone plus drop cables/terminators, so it's cheaper for small networks.

Reliability — correct, with one nuance: in star topology, if the *central device* (hub/switch) fails, the entire network goes down, not just one node. So star isolates *node* failures well, but it creates a single point of failure at the hub — worth mentioning if this is for an exam or assignment, since it's a common follow-up question.

Performance — mostly correct, but the reasoning depends on whether the central device is a hub or a switch:

- A **hub** just repeats signals to all ports, so it doesn't actually reduce collisions — you still get a shared collision domain, similar to bus.
- A **switch** is what actually reduces collisions, because it forwards frames only to the intended port and gives each connection its own collision domain.

So the stronger statement is: "star topology *using a switch* offers better performance," rather than star topology in general.

One thing to consider adding: installation/maintenance ease — bus is simpler to set up initially but harder to troubleshoot (a single break is hard to locate and affects everyone), while star is easier to diagnose and expand (just add another node to the switch) even though initial wiring is more labor-intensive.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Answer: Mesh's redundancy (multiple paths per node) vs. star's single point of failure at the hub is the right comparison, and the conclusion about mesh being more robust in critical environments is well-supported.

A few refinements worth considering:

Distinguish full vs. partial mesh — this matters a lot for accuracy:

- **Full mesh:** every node connects to every other node directly. Maximum fault tolerance, but connections grow as $n(n-1)/2$ — expensive and impractical beyond a small number of nodes.
- **Partial mesh:** only some nodes have multiple connections; others may have just one or two. This is far more common in practice (e.g., WAN backbones, some ISP networks) and offers *good* but not *maximum* fault tolerance — a compromise between full mesh and cheaper topologies.

If your point 1 is meant as a general statement about mesh, it's more precise to say "full mesh offers very high fault tolerance" since partial mesh's fault tolerance depends on how many redundant links actually exist.

Cost/complexity tradeoff — worth pairing with the reliability point, since it's the natural counterbalance: mesh's high fault tolerance comes at the cost of expensive cabling, complex configuration, and difficult scalability, which is exactly why it's reserved for critical environments (military networks, core internet infrastructure, some industrial control systems) rather than used generally.

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Answer:

1. Installation Complexity: Bus vs. Mesh

Bus topology is simple and easy to install because it uses a single backbone cable, with all devices connected to it via drop lines or connectors along its length. There's no need for a central device, and adding a new node typically just means tapping into the existing cable, making setup fast and requiring minimal technical expertise. This is why bus topology has historically been favored for small networks, temporary setups, or environments with tight budgets and simple requirements.

Mesh topology, by contrast, is highly complex to install because it requires a dedicated connection between every pair of nodes (in a full mesh). This means the number of cables and ports needed grows very quickly—following the formula $n(n-1)/2$, where n is the number of nodes. For example, a 5-node network needs only 10 connections, but a 10-node network needs 45, and a 20-node network needs 190. Each node also needs enough network interfaces/ports to support all its connections, which adds hardware cost and configuration overhead on top of the cabling itself.

As a result, the installation effort for mesh topology scales poorly—what's manageable for a handful of devices becomes impractical and extremely costly for larger networks. This is why mesh is generally reserved for small critical cores (e.g., router backbones, WAN links between major sites) rather than general-purpose LANs, whereas bus topology's simplicity made it practical for basic small-scale networks, even though it has since been largely replaced by star topology in modern networks due to its own drawbacks (like the single cable being a point of failure).

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on

the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Answer:

1. Scalability: Hybrid vs. Star Topology

Hybrid topology is highly scalable because it combines two or more different topologies—such as star, mesh, bus, or ring—allowing network designers to choose the best characteristics of each for different parts of the network. For example, a large organization might use star topology within individual departments (for easy management and fault isolation) while linking those department networks together using a bus or mesh backbone. This modular approach means new segments, buildings, or departments can be added as independent units without redesigning or disrupting the existing network structure—you simply extend the backbone or add another star cluster and connect it in.

This flexibility also means hybrid networks can be tailored to specific needs at each level: a company might use mesh topology for its critical server core (where fault tolerance matters most), star topology for end-user workstations (where ease of management matters most), and bus topology for a small legacy segment that doesn't need upgrading. Because expansion happens at the segment level rather than requiring changes to the whole network, hybrid topology can grow to accommodate large, geographically distributed, or organizationally complex networks—such as university campuses, corporate headquarters with multiple branches, or citywide networks—far more effectively than a single topology could on its own.

Star topology, in comparison, is also scalable, but only within limits set by the central device. Adding a new node is as easy as long as the switch or hub has a free port, but once the device reaches its maximum port capacity, further expansion requires purchasing and installing additional switches, then interconnecting them (often via uplink ports or cascading). This adds cost, introduces additional points of potential failure, and can increase latency if not designed carefully, since traffic between devices on different switches must now pass through the interconnection. Furthermore, in a very large star network, the central switch(es) can become a performance bottleneck, since all traffic must pass through them.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Answer:

Star—correct. Since each node has its own dedicated link back to the central device, a fault shows up as an isolated cable or port problem, easy to trace using the switch's diagnostic tools/port lights.

Mesh—correct. With many redundant links per node, tracing which specific connection is faulty is harder, and network monitoring tools become essential rather than optional. Worth noting: mesh's *fault tolerance* actually works against easy maintenance in one sense—because alternate paths keep the network running, a failed link can go unnoticed longer, delaying repair unless proactive monitoring is in place.

Bus—correct, and this is one of bus topology's most cited real-world weaknesses: a break or loose connector anywhere along the backbone can bring down the whole segment, and

technicians often have to test cable segments sequentially (or use a cable tester/reflectometer) to find the fault point, since there's no central device reporting per-node status.

Hybrid — correct. Worth adding: the maintenance difficulty in hybrid networks often depends on documentation and network design quality — a well-documented hybrid network (clear diagrams of which segments use which topology) is manageable, but a poorly planned one can become genuinely difficult to troubleshoot since faults could originate in any of the combined topology types, each with different failure characteristics.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary(4)	Proficient(3)	Developing(2)	Beginning(1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion